

Main Ideas

Copy and complete these statements to make a summary of this chapter. The missing words are on the right.

- _____ are found in the nuclei of cells. They carry _____ which determine what an organism looks like and how it functions.
- Body cells contain pairs of chromosomes and _____ contain only single chromosomes. So when fertilisation occurs, the single chromosomes in the sex cells form pairs in the cells of the new organism.
- Sex in humans is determined by sex chromosomes—females have a pair of _____ and males have an X and a _____.
- Organisms carry two genes for each characteristic. Some genes are _____ and mask the effects of the other (recessive) genes.
- The _____ and phenotypes of offspring can be predicted using _____.
- Chromosomes are made of _____. The sequence of the _____ on the DNA determines which types of proteins will be made by the cell.
- _____ alter the genetic material in cells, and occur spontaneously or from exposure to radiation or certain chemicals.
- Scientists throughout the world have collaborated to _____ all the genes in the human _____.

bases
chromosomes
DNA
dominant
genes
genome
genotypes
map
mutations
Punnett squares
sex cells
X chromosomes
Y chromosome

Test Yourself

- Chromosomes are found in—
 - sex cells only
 - all cells with a nucleus
 - fertilised eggs only
 - animal cells only
- A gene is—
 - a chromosome
 - a molecule of DNA
 - part of a chromosome that carries a single instruction
 - one base on a molecule of DNA
- A guinea pig has brown hair. Which one of the following statements is correct?
 - Both genes for hair colour came from the male parent.
 - Both genes came from the female parent.
 - One gene came from each parent.
 - Two genes came from each parent.
- Horses have a total of 64 chromosomes in each of their body cells. Male horses have an X and a Y chromosome in their cells.
 - How many pairs of chromosomes are found in horses?
 - How many chromosomes are found in the sperm of a horse?
 - Which chromosomes do sperm carry?

Review Questions

1. Complete the table to compare mitosis and meiosis in humans.

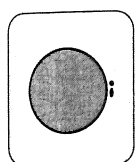
	MITOSIS	MEIOSIS
Function		
Location		
Chromosome number in parent cell		
Chromosome number in daughter cells		
Number of daughter cells		
Number of divisions		

2. (i) When meiosis occurs in the testes what proportion of the gametes formed contain an X chromosome?
- _____
- (ii) In the testes what proportion of gametes formed contain a Y chromosome?
- _____
- (iii) When meiosis occurs in the ovaries what proportion of the gametes contain an X chromosome?
- _____
- (iv) How does this affect the proportion of males and females in the population?
- _____

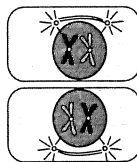
(iii) Define diploid.

(iv) Which cells are diploid?

2. (i) The following sequence shows a cell undergoing meiosis. This is not a human cell. In the spaces provided, describe what is occurring in and to the cell.



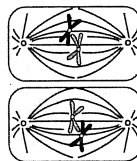
1. _____



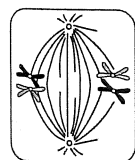
5. _____



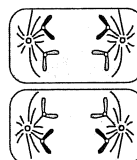
2. _____



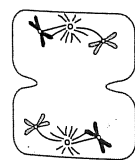
6. _____



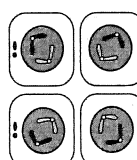
3. _____



7. _____



4. _____



8. _____

In the example above:

- (ii) How many chromosomes did the parent cell contain? _____
- (iii) How many chromosomes do each of the gametes contain? _____
- (iv) What is the total number of chromosomes in the four daughter cells? _____
- (v) How do you account for the apparent increase in the total number of chromosomes? _____

12.2 MEIOSIS



Terminology

These are some of the terms from this section which you should know. Write the meaning of each term in the space provided.

(i) cell division

(ii) chromatid

(iii) chromatin

(iv) cytokinesis

(v) daughter cell

(vi) homologous chromosomes

Review Questions

1. (i) Define **haploid**.

(ii) Which cells are haploid?
